

30 CHAPTERWISE IMPORTANT QUESTIONS

Science (086) | CBSE 2026-27

How Do Organisms Reproduce?

CBSE Science 2026-27 | Board-examinable

30 original revision questions with answers: short-answer, competency, assertion-reason and long-answer practice. Set is based on the current syllabus and official question-paper style; it is not a prediction or an official CBSE paper.

Question 1

Why is DNA copying essential in reproduction, and why is it not perfectly exact?

Answer:

DNA copying passes hereditary information to offspring. Small copying differences create variation, which may help a population survive environmental change; exact copying alone would reduce diversity.

Question 2

Compare binary fission and budding with one example each.

Answer:

In binary fission one parent cell divides into two similar individuals, as in Amoeba. In budding a small outgrowth develops and separates, as in yeast or Hydra.

Question 3

What is the role of the placenta during human pregnancy?

Answer:

The placenta allows exchange of nutrients and oxygen from the mother's blood to the embryo and carries wastes away. It connects maternal and foetal circulation without normally mixing their blood directly.

Question 4

What is the central idea or skill tested in How Do Organisms Reproduce??

Answer:

The central focus is: DNA copying passes hereditary information to offspring. This is the starting point for a strong answer from this chapter.

Question 5

Write two relevant points for a short answer based on How Do Organisms Reproduce?.

Answer:

First point: DNA copying passes hereditary information to offspring. Second point: In binary fission one parent cell divides into two similar individuals, as in Amoeba. Link the two points clearly instead of listing them without explanation.

Question 6

How does the first key idea in How Do Organisms Reproduce? lead to an important result or conclusion?

Answer:

Begin with the key idea: DNA copying passes hereditary information to offspring. Then connect it to the result: In binary fission one parent cell divides into two similar individuals, as in Amoeba.

How Do Organisms Reproduce? - Continued

Question 7

What evidence, calculation, incident or concept best supports the first main idea in How Do Organisms Reproduce??

Answer:

Use this supporting detail: DNA copying passes hereditary information to offspring. Small copying differences create variation, which may help a population survive environmental change; exact copying alone would reduce diversity. The relevant detail should be stated before giving the final conclusion.

Question 8

Explain the importance of the second major idea in How Do Organisms Reproduce?.

Answer:

In binary fission one parent cell divides into two similar individuals, as in Amoeba. In budding a small outgrowth develops and separates, as in yeast or Hydra. Its importance lies in showing how the chapter develops the topic beyond a simple fact or definition.

Question 9

What common mistake should a student avoid while answering a question from How Do Organisms Reproduce??

Answer:

Do not give only a conclusion. Explain the reasoning with a correct detail, such as: The placenta allows exchange of nutrients and oxygen from the mother's blood to the embryo and carries wastes away.

Question 10

Compare the first and second important ideas in How Do Organisms Reproduce?.

Answer:

The first idea is: DNA copying passes hereditary information to offspring. The second idea is: In binary fission one parent cell divides into two similar individuals, as in Amoeba. A complete comparison states both the difference and the connection between them.

Question 11

Show a cause-and-effect link from How Do Organisms Reproduce?.

Answer:

Cause or starting point: DNA copying passes hereditary information to offspring. Effect or outcome: In binary fission one parent cell divides into two similar individuals, as in Amoeba. Use connecting words such as 'therefore', 'as a result', or 'thus'.

Question 12

What is the most useful sequence for explaining the main content of How Do Organisms Reproduce??

Answer:

A clear sequence is: DNA copying passes hereditary information to offspring. Next, In binary fission one parent cell divides into two similar individuals, as in Amoeba. Finally, The placenta allows exchange of nutrients and oxygen from the mother's blood to the embryo and carries wastes away.

How Do Organisms Reproduce? - Continued

Question 13

Give a 40-50 word answer outline for an important question from How Do Organisms Reproduce?.

Answer:

Start with the direct answer: DNA copying passes hereditary information to offspring. Add one supporting point: In binary fission one parent cell divides into two similar individuals, as in Amoeba. End with the outcome or message: The placenta allows exchange of nutrients and oxygen from the mother's blood to the embryo and carries wastes away.

Question 14

Which scientific observation, equation, diagram, or principle should be written accurately in How Do Organisms Reproduce??

Answer:

Use an accurate scientific statement: DNA copying passes hereditary information to offspring. Then link it to the observation, process, or result asked in the question.

Question 15

How can an examiner see that you understand, rather than memorise, How Do Organisms Reproduce??

Answer:

Connect the idea to its reason or result. For instance: In binary fission one parent cell divides into two similar individuals, as in Amoeba. Then explain why this point matters in the chapter's context.

Question 16

State one inference that can be drawn from the main situation or concept in How Do Organisms Reproduce?.

Answer:

A sound inference is that the placenta allows exchange of nutrients and oxygen from the mother's blood to the embryo and carries wastes away. This inference must be supported by a relevant point from the chapter.

Question 17

Frame a short competency-based response using the first two ideas in How Do Organisms Reproduce?.

Answer:

Apply the concept by first stating: DNA copying passes hereditary information to offspring. Then justify the response with: In binary fission one parent cell divides into two similar individuals, as in Amoeba.

Question 18

What would make an answer from How Do Organisms Reproduce? incomplete?

Answer:

An answer is incomplete if it omits the explanation or final implication. Include both: DNA copying passes hereditary information to offspring. Also add: The placenta allows exchange of nutrients and oxygen from the mother's blood to the embryo and carries wastes away.

How Do Organisms Reproduce? - Continued

Question 19

Prepare an Assertion-Reason style explanation based on How Do Organisms Reproduce?.

Answer:

Assertion: DNA copying passes hereditary information to offspring. Reason: In binary fission one parent cell divides into two similar individuals, as in Amoeba. Check whether the reason actually explains the assertion before choosing an option.

Question 20

How should you justify an answer from How Do Organisms Reproduce? in two clear steps?

Answer:

Step 1: state the concept or event - DNA copying passes hereditary information to offspring. Step 2: support it with the explanation or result - In binary fission one parent cell divides into two similar individuals, as in Amoeba.

Question 21

What answer structure is suitable for a 3-mark question from How Do Organisms Reproduce??

Answer:

Use three connected points: DNA copying passes hereditary information to offspring. In binary fission one parent cell divides into two similar individuals, as in Amoeba. The placenta allows exchange of nutrients and oxygen from the mother's blood to the embryo and carries wastes away. Keep the points specific and write them in a logical order.

Question 22

What answer structure is suitable for a 5-mark question from How Do Organisms Reproduce??

Answer:

Give an introduction, develop these key points - DNA copying passes hereditary information to offspring. In binary fission one parent cell divides into two similar individuals, as in Amoeba. - and close with: The placenta allows exchange of nutrients and oxygen from the mother's blood to the embryo and carries wastes away. Add a diagram, equation, example, or textual reference where relevant.

Question 23

How can the ideas in How Do Organisms Reproduce? be applied to a new situation or case study?

Answer:

Identify the relevant principle first: DNA copying passes hereditary information to offspring. Then use the chapter's reasoning to explain the case: In binary fission one parent cell divides into two similar individuals, as in Amoeba.

How Do Organisms Reproduce? - Continued

Question 24

Which detail from How Do Organisms Reproduce? is most likely to earn an extra mark when explained correctly?

Answer:

A useful detail is: The placenta allows exchange of nutrients and oxygen from the mother's blood to the embryo and carries wastes away. Explain it in context instead of using it as an isolated statement.

Question 25

Write a brief concluding statement for an answer on How Do Organisms Reproduce?.

Answer:

A suitable conclusion is: The placenta allows exchange of nutrients and oxygen from the mother's blood to the embryo and carries wastes away. This conclusion should follow naturally from the explanation rather than introduce a new point.

Question 26

Make a three-point revision note for How Do Organisms Reproduce?.

Answer:

1. DNA copying passes hereditary information to offspring. 2. In binary fission one parent cell divides into two similar individuals, as in Amoeba. 3. The placenta allows exchange of nutrients and oxygen from the mother's blood to the embryo and carries wastes away. Revise these points with the chapter's examples, terms, and practice questions.

Question 27

How would you self-check an answer from How Do Organisms Reproduce? before submitting it?

Answer:

Check that you have answered the exact question, used the key idea - DNA copying passes hereditary information to offspring. - and supported it with a reason, example, calculation, or outcome such as: In binary fission one parent cell divides into two similar individuals, as in Amoeba.

Question 28

Why does How Do Organisms Reproduce? remain important for revision?

Answer:

It combines factual recall with explanation and application. The main takeaways are: DNA copying passes hereditary information to offspring. In binary fission one parent cell divides into two similar individuals, as in Amoeba. The placenta allows exchange of nutrients and oxygen from the mother's blood to the embryo and carries wastes away.

Question 29

Create a final 60-80 word answer plan for a long question from How Do Organisms Reproduce?.

Answer:

Opening: DNA copying passes hereditary information to offspring. Development: In binary fission one parent cell divides into two similar individuals, as in Amoeba. Conclusion: The placenta allows exchange of nutrients and oxygen from the mother's blood to the embryo and carries wastes away. Use complete sentences and include all required evidence, steps, or references.

How Do Organisms Reproduce? - Continued

Question 30

What is one last-minute revision question you should ask yourself about How Do Organisms Reproduce??

Answer:

Ask: 'Can I explain why this is true or important?' A correct self-answer should include: DNA copying passes hereditary information to offspring. and In binary fission one parent cell divides into two similar individuals, as in Amoeba.

Sources: [CBSE Science curriculum 2026-27](#); [CBSE Class X sample papers and marking schemes 2026-27](#); [NCERT textbooks](#).

ONE SHOT MASTERY